

coming more important. The two are focusing on manufacturing long-haul heavy freight electric locomotives to run on the dedicated freight corridor. Alstom is providing 100 high-power electric locomotives annually to the Railways for freight service. Siemens will provide 1,200 locomotives over 11 years; the first was delivered in May his year.

India has around 4,397 diesel locomotives. The Railways has decided to keep 2,500 for disaster management and strategic purpose. Only 1,000 locomotives have been assigned to continue operations. Also, every year, 2-3% locomotives enter the end of life, which means that of the total locomotive strength of around 15,000, 300-450 go out of the system. The lifespan of a locomotive is around 35 years.

Sanjiv Garg, former Additional Member, Railway Board, says the railway ministry took a decision in 2018 for 100% electrification as part its 2030 Net Zero target and decided to replace all diesel engines with electric.

"It was a flawed decision. The idea should have been to electrify the high density network rather than branch lines. The cost of overhead equipment (OHE) is ₹2 crore per km. Now, in case of a natural calamity or train accident, if the OHE network is damaged, the entire railway network will be impacted. In such situations, diesel locomotives are required for immediate relief and restoration purposes," says Garg, now Secretary General of the Chartered Institute of Logistics and Transport, which represents transport and logistics industries worldwide.

Garg questions how thermal-powered electric locomotives can be greener than diesel locomotives. "Both are powered by fossil fuels," he says, adding that diesel locomotives should have remained functional on

BY THE NUMBERS



13,198

**PASSENGER
TRAINS**



11,724

GOODS TRAINS



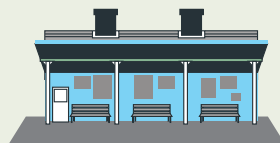
70,000

**KILOMETRE
ROUTE LENGTH**



15,110

LOCOMOTIVES



7,461

STATIONS

the branch network. "Condemnation of productive diesel locomotives prematurely means wastage of a high-cost resource, as some of the diesel engines still have 15-20 years operational life. Railways should have gradually phased them out and slowly ramped up electric locomotive production as per requirement," he says. Several countries still run diesel locomotives. The Railways is repurposing the condemned diesel locomotives for countries in Africa after gauge conversion.

Interestingly, Russia-Ukraine and Israel-Iran conflicts have made countries realise the importance of retaining diesel locomotives for contingencies as power grids are a prime target in any war. Ukraine, engaged in war with Russia, has ordered 225 diesel locomotives and the overhauling of 75 existing locomotives from US-based Watbtec (former GE Transportation).

The decision, announced in April this year, came in the wake of Russia's repeated attacks on Ukraine's power grid, impacting railways. In 2020, Ukraine had decided to completely switch to electric locomotives.

DIESEL CHALLENGE

The move towards electrification has forced companies to tweak their plans. Watbtec is focusing on the 'Make in India for Export' initiative, with the first two locomotives headed to Guinea in West Africa after being flagged off by Prime Minister Narendra Modi recently.

Under the 'Make in India' programme, Watbtec's plant in Marhowra, Bihar, was tasked with supplying 1,000 fuel-efficient, emission-compliant diesel-electric Evolution Series locomotives as part of a \$2.5-billion public private partnership.